

XI LI

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Assistant Professor, Department of Computer Science, University of Alabama at Birmingham

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RESEARCH INTERESTS

Multimodal AI Safety, Adversarial Machine Learning, Trustworthy AI, AI Agents

My research develops methods to identify, understand, and mitigate AI failures, enabling safe and reliable AI deployment in healthcare, sustainable infrastructure, and other high-impact domains.

ACADEMIC APPOINTMENTS

Tenure-Track Assistant Professor

Aug. 2024 – present

Department of Computer Science, UAB

Associate Scientist

2025 – present

Integrative Center for Aging Research, School of Medicine, UAB

Center for the Study of Community Health, School of Public Health, UAB

EDUCATION

The Pennsylvania State University

State College, PA

Ph.D. in Computer Science and Engineering

Aug. 2018 - Aug. 2024

Advisor: Dr. George Kesidis and Dr. David Miller

M.S. in Computer Science and Engineering

Aug. 2016 - July 2018

The Southeast University

Nanjing, China

B.E. in Electrical Engineering

Aug. 2012 - June 2016

GRANTS

Funded Projects

1. Collaborative Research: Elements: AI-Agent Digital Twin for Collaborative Building Auditing and Retrofit Simulation.

NSF, Award No. 2609220, 2026–2029.

Role: UAB PI. Total Amount: \$600,000. UAB Share: \$299,997.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern

May 2024 – Aug. 2024

Modern Recommendation System Team, Meta

New York

PUBLICATIONS

Conference Papers

C.1 Yuqiao Meng, Luoxi Tang, Feiyang Yu, **Xi Li**, Guanhua Yan, Ping Yang, Zhaohan Xi. Benchmarking LLM-Assisted Blue Teaming via Standardized Threat Hunting. *ICML 2026*.

- C.2 **Xi Li**, Ruofan Mao, Yusen Zhang, Renze Lou, Chen Wu, Jiaqi Wang. Chain-of-Scrutiny: Detecting Backdoor Attacks for Large Language Models. *ACL 2025 (Findings)*.
- C.3 Falong Fan, **Xi Li**. PeerGuard: Defending Multi-Agent Systems Against Backdoor Attacks Through Mutual Reasoning. *IEEE IRI 2025. Best Paper Award*.
- C.4 Xiaohuan Bi, **Xi Li**. Securing Federated Learning Against Novel and Classic Backdoor Threats During Foundation Model Integration. *IEEE IRI 2025*.
- C.5 Renze Lou, Hanzi Xu, Sijia Wang, Jiangshu Du, Ryo Kamoi, Xiaoxin Lu, Jian Xie, Yuxuan Sun, Yusen Zhang, Jihyun Janice Ahn, Hongchao Fang, Zhuoyang Zou, Wenchao Ma, **Xi Li**, Kai Zhang, Congying Xia, Lifu Huang, Wenpeng Yin. AAAR-1.0: Assessing AI's Potential to Assist Research. *ICML 2025; Best Paper Award at AAAI 2025 AI4Research workshop*.
- C.6 Guangmingmei Yang, **Xi Li**, Hang Wang, George Kesidis, David J Miller. Backdoor Inversion in Neural-Activation Space. *MLSP 2025*.
- C.7 Fei Zhao, Chengcui Zhang, Runlin Zhang, Tianyang Wang, **Xi Li**. Mitigating Image Captioning Hallucinations in Vision-Language Models. *IEEE MIPR 2025*.
- C.8 **Xi Li**, Chen Wu, Jiaqi Wang. Vulnerabilities of Foundation Model Integrated Federated Learning Systems Under Adversarial Threats. *IJCNN 2025*.
- C.9 **Xi Li**, Chen Wu, Jiaqi Wang. Unveiling Backdoor Risks Brought by Foundation Models in Heterogeneous Federated Learning. *PAKDD 2024*.
- C.10 **Xi Li**, Songhe Wang, Ruiquan Huang, Mahanth Gowda, George Kesidis. Temporal-Distributed Backdoor Attack Against Video Based Action Recognition. *AAAI 2024*.
- C.11 **Xi Li**, David J Miller, Zhen Xiang, George Kesidis. A BIC-based Mixture Model Defense against Data Poisoning Attacks on Classifiers. *IEEE MLSP 2023*.
- C.12 **Xi Li**, David J Miller, Zhen Xiang, George Kesidis. Test-Time Detection of Backdoor Triggers of Poisoned Deep Neural Networks. *ICASSP 2022*.
- C.13 Zhen Xiang, David J Miller, Siheng Chen, **Xi Li**, George Kesidis. Detecting Backdoor Attacks Against Point Cloud Classifiers. *ICASSP 2022*.
- C.14 Zhen Xiang, David J Miller, Siheng Chen, **Xi Li**, George Kesidis. A Backdoor Attack against 3D Point Cloud Classifiers. *ICCV 2021*.

Journal Articles

- J.1 Xi Xiao, Xingjian Li, Cheng Han, Tianyang Wang, Lin Zhao, Yunbei Zhang, Guosheng Hu, Runmin Jiang, **Xi Li**, Xiao Wang, Min Xu. Adapting Vision Foundation Models with Cascaded Semantics. *Transactions on Machine Learning Research (TMLR)*, 2026.
- J.2 **Xi Li**, Zhen Xiang, David J Miller, George Kesidis. Correcting the distribution of batch normalization signals for Trojan mitigation. *Neurocomputing*, 2024.
- J.3 **Xi Li**, David J Miller, Zhen Xiang, George Kesidis. BIC-based Mixture Model Defense against Data Poisoning Attacks on Classifiers: A Comprehensive Study. *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 2024.

Workshop Papers

- W.1 **Xi Li**, Shu Zhao, Xiaohan Zou, Fei Zhao, Fuxiao Liu, Yusen Zhang, Cheng Han, Yushun Dong,

- Jiaqi Wang. Evolving Safety Landscape of Multi-modal Large Language Models: A Survey of Emerging Threats and Safeguards. *ICLR Workshop on Principled Design for Trustworthy AI*, 2026.
- W.2** Xi Li, Shu Zhao, Fei Zhao, Runlong Yu. Rethinking the Safety Landscape for Foundation Models: A Multi-Modal Perspective. *ICCV T2FM*, 2025.
- W.3** Xi Li, Jiaqi Wang. Federated Foundation Models Raise New Concerns of Robustness, Privacy, and Fairness. *ICCV T2FM*, 2025.
- W.4** Xi Li, Songhe Wang, Chen Wu, Hao Zhou, Jiaqi Wang. Backdoor Threats from Compromised Foundation Models to Federated Learning. *FL@FM-NeurIPS'23*.

Preprints

- P.1** Shirong Kou, Shu Zhao, Cheng Han, Zhuowen Tu, Xi Li. One Prompt across Modalities: Unified Robustness and Safety Adaptation for Vision-Language Models. under review, 2026.
- P.2** Falong Fan, Qizhen Lan, Xiangchen Li, Xi Li, Miaomiao Zhang, Xiaoqian Jiang. No Visit Is Removable: A Counterfactually Certified Benchmark for Multi-Visit Chest-Radiograph Reasoning. under review, 2026.
- P.3** Xi Li, Shu Zhao, Xiaohan Zou, Fuxiao Liu, Fei Zhao, Falong Fan, Arun Zachariah, Yusen Zhang, Cheng Han, Yushun Dong, Jiaqi Wang, Bo Liu, Chengcui Zhang, Zhuowen Tu. Evolving Safety Landscape of Multi-modal Large Language Models: A Survey of Emerging Threats and Safeguards. under review, 2026.
- P.4** Chenjia Li, Ruiquan Huang, Yusen Zhang, Lingjuan Lyu, Xi Li, Jiaqi Wang. ParaGen-Bench: A Benchmark for Neural Network Parameter Generation. under review, 2026.
- P.5** Huan Ma, Yiyang Liu, James Chenhao Liang, Qihao Ye, Chole Lou, Xi Li, Dongfang Liu, Qifan Wang, Tianyang Wang, Cheng Han. SaRA: Sink-aware Representation Allocation for Restoring Visual Semantics in Multimodal LLMs. under review, 2026.
- P.6** Jiaqi Wang, Yusen Zhang, Xi Li, Simeng Han, Zining Zhu, Lingjuan Lyu, Yaqing Wang, Ruiquan Huang, Arman Cohan, Rui Zhang, Sheng Li, Xin Wang. Survey on Neural Network Parameter Acquisition: from Optimization to Generation. under review, 2026.
- P.7** Jiaqi Wang, Yusen Zhang, Xi Li. NeuroGen: Neural Network Parameter Generation via Large Language Models. under review, 2025.

TUTORIALS

- Towards Safe Multi-Modal Learning: Evolving Threats and Safety Solutions, *CVPR 2026*.
- Towards Safe Multi-Modal Learning: Unique Challenges and Future Directions, *ICCV 2025*.
- Exploitation and Mitigation: Understanding Large-Scale Machine Learning Robustness under Paradigm Shift, *SDM 2025*.

INVITED TALKS

- The Evolving Safety Landscape of Multi-Modal LLMs, *AI@AU Forum, Auburn University, 2026*.
- Towards Safe Multi-Modal Learning: Unique Challenges and Future Directions, *School of Computing, University of Georgia, 2025*.

- Towards Trustworthy AI: Evolving from Secure Systems to Responsible Integration, *School of Electrical Engineering and Computer Science, Ohio University, 2024.*

AWARDS

- **Best Paper Award**, *IEEE IRI, 2025.*
- **Best Paper Award**, *AI4Research Workshop @ AAAI 2025*

TEACHING EXPERIENCE

- **Instructor** UAB
 - CS652/752 Advanced Algorithms and Applications 2025-2026 Spring
 - CS685/785 Foundation of Data Science 2024-2026 Fall
 - CS300/500 Advanced OO Programming in C++ 2025-2026 Fall
- **Teaching Assistant** PSU
 - CMPSC465 Data Structures and Algorithms 2021 Spring
 - CMPSC/MATH 451 Numerical Computations 2019 Fall - 2024 Spring
 - CMPEN/EE 455 Digital Image Processing I 2018 Fall

MENTORING

Ph.D. Students

- **Shirong Kou**, Computer Science, UAB 2025–present
Research: Multimodal AI safety and adversarial machine learning.
- **Peiyuan Zhang**, Computer Science, UAB 2025–present
Research: Trustworthy AI for time-series data.

Master's Students

- **Falong Fan**, M.S. Data Science, CUHK-Shenzhen 2024-2025
Now Ph.D. student, University of Virginia.
- **Ruofan Mao**, Computer Science, UAB 2025
Now Data Information Coordinator, Biomedical Informatics and Data Science, UAB.

Undergraduate Student Researchers

- **Xiaohuan Bi**, Statistics, Renmin University of China 2024

Ph.D. Dissertation Committee Member

- **Neethu Suma Raveendran**, Mathematics, UAB 2026–present
- **Xi Xiao**, Computer Science, UAB 2026–present

M.S. Thesis Committee Member

- **Chen Liu**, Computer Science, UAB 2026–present

PROFESSIONAL SERVICE

- **Proposal Panelist:** NSF SaTC, NAIRR
- **Program Committee Member:** AAAI, SDM, CIKM
- **Conference Reviewer:** ICML, ICLR, NeurIPS, ACM CCS, ICCV, ECCV, CVPR, AAAI, AISTATS, KDD, COLM, ICASSP
- **Journal Reviewer:** IEEE Transactions on Neural Networks and Learning Systems (TNNLS), Neurocomputing, Neural Networks, Computers & Security, IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), IEEE Transactions on Mobile Computing (TMC)
- **Student Volunteer:** NeurIPS